

# SHUVAM PAL

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## Education

**Dayananda Sagar College of Engineering**

Bengaluru, India

*B.E. Artificial Intelligence and Machine Learning*

## Experience

**Bentham AI** 

Dec 2025 – Feb 2026

*Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions*

Remote

- Architected a resilient **Puppeteer/Node.js browser automation engine** for MCA filings, reducing manual filing time by **85%** through network-idle monitoring, CAPTCHA handoff, retries, and state recovery.
- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across MCA workflow preparation.
- Containerized services with **Docker**, deployed to **Google Cloud Run**, and automated releases through **GitHub Actions**, improving deployment velocity by **70%**.

**Google Summer of Code** 

May 2025 – Sept 2025

*Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX*

Remote

- Built MRI segmentation tooling for **95 anatomical regions**, integrating model inference outputs into existing visualization and user-interaction paths.
- Reduced processing time by **87%** using async and parallel execution over expensive model and image-processing workflows.
- Enhanced 3D surface visualization so generated masks and model outputs remained inspectable in the application.

## Open Source

[Contributions](#) 

- Contributed to Invesalio, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working within existing codebases and maintainer review loops.

## Projects

**Postificus** | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, AWS, Prometheus*

[GitHub](#) 

- Built a **content distribution platform** using Go-Rod browser automation to cross-post to Medium, Dev.to, and Hashnode when direct publishing APIs were insufficient.
- Engineered RabbitMQ workers with DLQs, retries, Redis-backed autosave, circuit breakers, Prometheus metrics, and health checks for long-running third-party browser tasks.

**Murdoc** | *Python, FastAPI, MCP, OpenTelemetry, Prometheus, Docker*

[GitHub](#) 

- Built an **AI gateway** for LLM, HTTP tool, and MCP traffic with route profiles, runtime policy checks, request scoring, and replayable decisions before downstream execution.
- Instrumented provider/tool flows with **OpenTelemetry** traces, Prometheus metrics, audit ledgers, and attack-lab validation so latency, failures, and unsafe actions could be investigated per request.

**Penny Lane Capital (Tradeage)** | *Python, Agno, Pydantic, SQLite, yfinance, JSONL*

- Built an **Agno Workflow** multi-agent research system with analyst roles, bull/bear debate, trader, risk review, and portfolio approval using Pydantic state, checkpoints, memory, and JSONL traces.
- Evaluated agent decisions with historical-market backtests, reward loops, validation harnesses, and **5 baselines**, producing return, drawdown, and win-rate scorecards across market windows.

## Technical Skills

**Languages:** Go, Python, TypeScript

**Frameworks:** Echo, FastAPI, Node.js, React

**Tools:** Go-Rod, Puppeteer, RabbitMQ, Redis, PostgreSQL, Docker, Prometheus, OpenTelemetry